

Felipe Moura de Carvalho

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Work Experience

Tecgraf Institute

Rio de Janeiro, RJ, Brazil

Software Engineer

February, 2020 - November, 2025

- Implemented and maintained features for a 3D digital twin software platform using agile methodologies.
- Developed algorithms to aid 3D data exchange from CAD softwares using OpenCascade Technology.
- Conducted research on Order-Independent Transparency (OIT).
- Conducted research on Efficient Font Rendering.

Technologies: C++, OpenGL, GLSL, C++/CLI, Python, Qt, CMake, Conan, Git, OpenCascade

University of Calgary

Calgary, AB, Canada

Postdoctoral Researcher - Rapid Reservoir Modelling Consortium

July, 2015 - June, 2019

- Conducted research in conjunction with geologist and reservoir engineer to investigate and develop new technologies to aid in the process of rapid concept 3D geological models using sketch based natural interfaces.
- Created the initial sketch-based interface and modelling infrastructure to kick off the project.
- Integrated the Flow Diagnostic reservoir simulator, and implemented the post-processing 3D visualization and the graphical user interface.
- Implemented a 3D geological mesh viewer to aid the modelling process using interactive illustrative visualization techniques.
- Launched and presented the prototype to the sponsors in the annual meetings.

Technologies: C++, CMake, Qt, OpenGL, GLSL, CGAL, Git

Education

Federal University of Rio de Janeiro

Rio de Janeiro, RJ, Brazil

Alberto Luiz Coimbra Institute for Graduate Studies and Research in Engineering

2010 - 2014

Ph.D. in System Engineering and Computer Science (Computer Graphics)

Thesis: [Interactive Cutaway Rendering of Corner-Point Models](#)

Federal University of Rio de Janeiro

Rio de Janeiro, RJ, Brazil

Alberto Luiz Coimbra Institute for Graduate Studies and Research in Engineering

2007 - 2010

M.Sc. in System Engineering and Computer Science (Computer Graphics)

Thesis: [Renderização de Pontos usando Multiresolução](#)

Federal University of Piauí

Teresina, PI, Brazil

Department of Computer Science and Statistics

2003 - 2007

B.Sc. in Computer Science

Final Project: *Inteligência Artificial usando Linguagem Haskell*

Languages

Portuguese (Native), English

Projects

Interactive Cutaways of Oil Reservoir

2012 - 2014

In this project, I have implemented an Interactive Cutaways method strongly relying on screen-space GPU techniques especially designed for inspecting 3D reservoir models represented as corner-point. Alongside, I also adapted some illustrative techniques to enhance spatial relationship among the internal primitives

Project Page: <https://doi.org/10.1016/j.gmod.2016.02.001>

The Digital Bamberg Project

2010 - 2021

The Digital Bamberg project was a joint collaboration between the Computer Graphics Lab at PESC/UFRJ, and the Museum of Astronomy and Related Sciences (MAST). It combined the challenges of creating a digital replica of a scientific instrument, that, at the same time, is part of the museum collection and, hence, is treated as an historical artifact. In the project, I have assisted development and design software and algorithms for the 3D digitization process, which include the acquisition of the 3D geometry, by a laser scanner, and the registration of high quality images, acquired by a professional DSLR camera, of a historic scientific instrument.

Project Page: <https://www.lcg.ufrj.br/bamberg/>

React Bar

2008

During the iMedia Workshop, I have implemented the software prototype *React Bar*. A collaborative effort in conjunction with a designer, as part of the workshop of Interactive Media and Natural Interfaces promoted by Visgraf at the Institute for Pure and Applied Mathematics. The workshop focused on topics related with recent advances in tangible user interfaces (TUI), multitouch interaction, computer graphics, image processing, sound synthesis and new authoring environments. The idea of *React Bar* is to use multitouch interaction and tangible interface for entertainment in public spaces, exploring the concept of social networking. It has a simple and intuitive interface, providing a fun user experience. It promotes real interaction among users, initiating interaction in a virtual environment.

Project Page: <https://www.visgraf.impa.br/imedia08/projects/reactbar.html>

Honor and Awards

The National Council for Scientific and Technological Development (CNPq) - Brazil Scholarship granted by CNPQ Brazil to spend one year at the University of Calgary as a visiting Ph.D Student (Grant no. 237044/2012-9)

Fundação de Amparo à Pesquisa do Estado do Rio de Janeiro - FAPERJ - March 2012 Bolsa Nota 10: Scholarship granted by FAPERJ with the purpose of rewarding the best graduate students in the state of Rio de Janeiro, Brazil. (Grant no. E-26/100.393/2012)

The National Council for Scientific and Technological Development (CNPq) - Brazil Scholarship granted by CNPQ Brazil for pursuing a Ph.D at COPPE/UFRJ (Grant no. 141232/2010-2)

The National Council for Scientific and Technological Development (CNPq) - Brazil Scholarship granted by CNPQ Brazil for pursuing a M.Sc at COPPE/UFRJ (Grant no. 132345/2007-2)

Open Source

contributed to *pymt* - multitouch framework for Python - <https://github.com/tito/pymt> It has evolved into *kivy* - <https://github.com/kivy>

contributed to *Tucano* - OpenGL wrapper and computer graphics library - <https://gitlab.com/lcg/tucano>

Social Media

linkedin	https://www.linkedin.com/in/felipe-decarvalho/
github	https://github.com/felipeCeler
orcid	https://orcid.org/0000-0002-3690-5480
google scholar	https://scholar.google.com/citations?user=AlsJB1cAAAAJ&hl=en
cnpq lattes	http://lattes.cnpq.br/0922142294203200 (in portuguese)

Technical Skills

Main Skills	computer graphics, illustrative visualization, digital twins, gpu programming, mesh processing, sketch-based interfaces, natural user interfaces, point-cloud processing, graphics applied to cultural heritage, HCI, image processing
Computer Programming Languages	cross-platform C++ development, Windows and Linux, Python, C#, C++/CLI, Haskell
Libraries/API/Software	OpenGL/GLSL, Qt, Eigen, CGAL, CMake, Conan, git, OpenVolumeMesh, OpenCascade, gRPC, NVIDIA CUDA, Meshlab, Blender, Vtk, Point Cloud Library, OpenCV

Invited Talks

Invited Speaker: CINCIT - Segundo Congreso Internacional de Computación e Innovación tecnológica. Andahuaylas - Apurímac - Perú. Title: <i>Smart Visibility Applied to Oil&Gas Reservoir Models</i>	Sept, 2019
Impa Seminar - Computer Graphics Rio de Janeiro, RJ, Brazil Title: <i>Renderização Interativa de Cutaways em Modelos de Reservatório de Petróleo (in portuguese)</i>	May, 2014
Impa Seminar - Computer Graphics Rio de Janeiro, RJ, Brazil Title: <i>Simplificação e Multiresolução em Modelos baseados em Pontos (in portuguese)</i>	April, 2010

Professional Service

Program Committee, Reviewer - IEEE Virtual Reality 2019

Research Visits

Computer Science Department, <i>illustrares Group</i> , University of Calgary, Canada Research on illustrative visualization of Oil and Gas reservoir models	Dec, 2012 - Dec, 2013
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Publications

Journal

Zhao Zhang, Sebastian Geiger, Margaret Rood, Carl Jacquemyn, Matthew Jackson, Gary Hampson, **Felipe Moura De Carvalho**, Clarissa Coda Marques Machado Silva, Julio Daniel Machado Silva, Mario Costa Sousa. [Fast flow computation methods on unstructured tetrahedral meshes for rapid reservoir modelling](#). *Computational GeoScience Journal*. v. 24, p. 641–661, 2020.

Zhao Zhang, Sebastian Geiger, Margaret Rood, Carl Jacquemyn, Matthew Jackson, Gary Hampson, **Felipe Moura De Carvalho**, Clarissa Coda Marques Machado Silva, Julio Daniel Machado Silva, Mario Costa Sousa. [A Tracing Algorithm for Flow Diagnostics on Fully Unstructured Grids With Multipoint Flux Approximation](#). *SPE JOURNAL*, v. 22, p. 1946-1962, 2017.

Felipe Moura de Carvalho, Emilio Vital Brazil, Ricardo Guerra Marroquim, Mário Costa Sousa, Antonio Oliveira, [Interactive cutaways of oil reservoirs](#). *Graphical Models*, v 84, p 1-14, 2016

MARROQUIM, Ricardo, Pfeiffer, Gustavo, **CARVALHO, F. M.**, Oliveira, Antonio A.F. [Texturing 3D models from sequential photos](#). *The Visual Computer*, v.28, p.1 - 11, 2012.

Conference Papers

Mario Costa Sousa, Julio Daniel Machado e Silva, Clarissa Coda Marques Machado, **Carvalho, Felipe Moura De**, Judice, Sicilia, Rahman, Fazilatur and Jacquemyn, Carl and Pataki, Margaret E. H. and Hampson, Gary J. and Jackson, Matthew D. and Petrovskyy, Dmytro and Geiger, Sebastian. [Smart Modelling of Geologic Stratigraphy Concepts using Sketches](#). Smart Tools and Apps for Graphics - Eurographics Italian Chapter Conference, 2020/11

Zhang Z, Geiger S, Rood M, Jacquemyn C, Jackson M, Hampson G, **de Carvalho FM***, Machado Silva CCM*, Machado Silva J*, Costa Sousa M.* (2018). [Fast Flow Computation Methods On Unstructured Tetrahedral Meshes For Rapid Reservoir Modelling](#). *ECMOR XVI - European Conf. on the Mathematics of Oil Recovery* Barcelona, Spain. 2018/09

Zhang Z, Geiger S, Rood M, Jacquemyn C, Jackson MD, Hampson GJ, **de Carvalho FM***, Machado Silva CC*, Silva JD*, Costa Sousa M.* (2017b). [Flow Diagnostics on Fully Unstructured Grids](#). *SPE Reservoir Simulation Conf. (RSC '17) - Session 12: Fast Simulation Methods and Next-Generation Simulator Development*. Montgomery, TX, USA, 2017/02.

MP Rood, M Jackson, G Hampson, EV Brazil, **F de Carvalho**, C Coda, MC Sousa, Z Zhang, S Geiger. [Sketch-based geologic modeling](#). *AGU Fall Meeting Abstracts*, 2015/12

MARROQUIM, Ricardo, PFEIFFER, G. T., **CARVALHO, F. M.**, OLIVEIRA, A. A. F. [Texturing 3D models with low geometric features](#) In: *XXVI Sibgrapi - Conference on Graphics, Patterns and Images*, 2011, Maceio. Graphics, Patterns and Images (Sibgrapi)., 2011. p.1 - 8

FILHO, Z., **CARVALHO, F. M.**, BRAZIL, E. A. V., MARROQUIM, Ricardo, SOUSA, M. C. [Cutaway Applied to Corner Point Models](#) In: *SIBGRAPI 2012 - WGARI (Workshop on Industry Applications)*, 2012, Ouro Preto. SIBGRAPI 2012 - WGARI (Workshop on Industry Applications)., 2012.

Poster with selection committee

MARROQUIM, Ricardo, **CARVALHO, F. M.**, DELLEPIANE, Matteo, OLIVEIRA, A. A. F. [3D Digitalization of Scientific Instruments The Case of the Bamberg Meridian Circle](#) In: *XXXI Symposium of the Scientific Instrument Commission*, 2012, Rio de Janeiro. Proceedings of the XXXI Symposium of the Scientific Instrument Commission.

André Maximo, Maria Paula Saba, Joana Passi, Ricardo Castañeda Marin, Fernando Ribeiro, Rodolfo Lima, Alexandra Alves, Patrik Matos, Erick Passos, Nina Paim, **Felipe Moura**, Luisa Fosco, Ilana Pateman, Marcelo Cicconet, and Luiz Velho, Interactive Media and Natural Interfaces In: [Electronic Language International Festival](#), 2009, São Paulo. FILE SYMPOSIUM., 2009

CARVALHO, F. M.; OLIVEIRA, A. A. F ; MARROQUIM, Ricardo . [Level of Detail for Point Model Rendering](#). In: XXI Brazilian Symposium on Computer Graphics and Image Processing, 2008, Campo Grande. Proceedings of the XXI SIBGRAPI, 2008.

